

Credit Card Performance Analysis

1. Problem Statement

A leading financial services company offering multiple credit card products (Silver, Gold, Platinum, Blue) is facing challenges in understanding its customer behavior, revenue drivers, and transaction patterns across different segments.

Despite having large volumes of transaction and customer data, the organization lacks a centralized analytical system to answer critical business questions such as:

- Which customer segments generate the most revenue?
- How spending behavior varies across income, age, and occupation?
- Which card categories and transaction types are most profitable?
- How acquisition cost impacts profitability?
- What factors influence customer engagement and transaction frequency?

To address these gaps, the company aims to leverage Business Intelligence (Power BI dashboards) to gain actionable insights into customer demographics, spending behavior, and transaction performance, enabling better strategic decisions in marketing, risk management, and revenue optimization.

2. Goal of the Dashboard

The goal of this dashboard is to provide a comprehensive analytical view of credit card business performance by combining customer and transaction data. It helps stakeholders identify high-value customer segments, analyze revenue contribution across different card categories, and track quarterly trends in transactions and income. The dashboard also enables a deeper understanding of customer demographics and spending behavior, evaluates transaction modes and expenditure patterns, and compares customer acquisition cost with revenue generation. Overall, it supports data-driven decision-making to improve profitability, optimize marketing strategies, and enhance customer satisfaction.

3. Key Visuals in Dashboard

Customer Dashboard

- KPI Cards (Revenue, Interest, Income, CSS)
- Revenue Trend by Gender (Line Chart)
- Age Group Distribution (Bar Chart)
- Customer Job Analysis (Table)
- Top States Contribution (Bar Chart)
- Salary Group Analysis
- Dependent Count Distribution
- Marital Status Breakdown

- Education Level Analysis
- Interactive Filters (Quarter, Card Type, Gender)

Transaction Dashboard

- KPI Cards (Revenue, Interest, Income, Transaction Count)
- Card Category Analysis (Table)
- Quarterly Revenue & Transaction Trend (Combo Chart)
- Revenue by Expenditure Type (Bar Chart)
- Revenue by Education (Bar Chart)
- Revenue by Customer Job (Bar Chart)
- Transaction Mode Analysis (Swipe/Chip/Online)
- Customer Acquisition Cost Analysis
- Filters (Quarter, Gender, Income Group, Card Type)

4. Insights (Based on Client Questions + Dashboard Data)

Customer Insights

- Total Revenue: 55M indicates strong business volume
- Male customers (~30M) contribute slightly higher revenue than females (~25M)
- Age Group 40–50 is the highest revenue contributor (~24M)
 - Middle-aged customers are the most valuable segment
- High-income group contributes the highest revenue (~29M)
 - Strong correlation between income and spending
- Top States (TX, NY, CA) contribute highest (~13M each)
 - Business is geographically concentrated
- Businessmen & White-collar jobs generate highest income and revenue
 - Premium segment drives profitability
- Married customers (~28M) spend more than single customers (~23M)
 - Family-oriented spending behavior
- Graduates contribute highest revenue (~22M)
 - Educated users show higher credit usage

Transaction Insights

- Blue card generates the highest revenue (~46M)
 - Most widely used card segment
- Total transactions: 656K → High usage frequency
- Q3 shows peak revenue (~14.2M)
 - Strong seasonal or campaign-driven growth
- Swipe transactions dominate (~35M revenue)
 - Physical transactions still dominant
- Top spending categories:
 - Bills (~14M)

- Entertainment (~10M)
 - Fuel (~9M)
- Customer acquisition cost highest for Blue card (~0.89M)
 - High acquisition but also high return
- Businessmen contribute highest transaction revenue (~17M)

5. Recommendations

1. Revenue Growth

- Focus on middle-aged (40–50) high-income customers
- Expand high-performing states (TX, NY, CA)

2. Product Strategy

- Promote Blue card (mass segment) for volume
- Upsell premium cards (Gold/Platinum) to high-income users

3. Customer Targeting

- Business professionals
- White-collar employees
- Married customers

4. Transaction Optimization

- Encourage digital payments (Chip/Online)
- Reduce dependency on swipe transactions

5. Cost Optimization

- Reduce customer acquisition cost via:
 - Digital marketing
 - Referral programs

6. Engagement Strategy

- Increase spending via:
 - Cashback offers
 - Category-based rewards (fuel, bills, travel)